

Cancelable Multimodal Biometric Authentication for Critical Infrastructure

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Abstract—Biometric traits cannot be reissued after a database breach, and multimodal systems multiply the sensitive data they retain, a serious concern for access control at critical-infrastructure facilities. This paper presents a multimodal verification system that combines face, voice, and a dynamic Z-shaped hand gesture and stores only keyed, revocable templates. Face and voice embeddings are mapped to 256-bit BioHash templates under HKDF-derived keys, and each enrollment is issued as a pool of four independently keyed template sets that can be revoked and replaced. The hand gesture is captured as MediaPipe landmark trajectories, matched with dynamic time warping, and protected by a keyed orthonormal transform. Presented modalities are combined under a default all-required fusion policy in which every factor must match individually. In single-reference tests on held-out LFW identities, protected face templates reach an equal error rate (EER) of 8.50%; on a closed-set VoxCeleb1 subset, the deployed two-sentence enrollment, one-sentence verification voice protocol reaches 1.27% EER. Revoked face and voice templates agree with their replacements only at chance level (Hamming similarity about 0.50, no acceptances). The untuned baseline thresholds still yield high false-rejection rates, the hand-gesture threshold has been calibrated only on synthetic data, and real-participant evaluation remains future work.

Index Terms—Cancelable biometrics, BioHashing, template protection, multimodal biometrics, speaker verification, face verification, hand-gesture biometrics, dynamic time warping.

I. INTRODUCTION

Access to critical infrastructure—power grids, industrial control rooms, government facilities—requires identity assurance stronger than knowledge-based credentials [1], [2]. Biometrics bind a credential to the person, but a trait cannot be changed: a leaked template is a permanent compromise and may link a person across applications [3], [4]. With a single modality, one bad capture or one attacked sensor also decides the outcome [5].

Multimodal biometrics reduce the dependence on any one trait [5], [6], and cancelable biometrics address the permanence problem by storing a keyed, repeatable transformation of the biometric instead of the biometric itself, so that a compromised template can be revoked and reissued under a new key [3], [4]. Combining them requires a protected

representation that preserves each modality's matching rule, a fusion rule in which a strong factor cannot compensate for a failed one, and one revocation lifecycle across all modalities.

This paper describes a web-based verification system with three complementary modalities: face, voice, and a dynamic in-air Z gesture. The gesture reuses the face camera and captures how a person moves rather than a static hand shape. The contributions are:

- 1) An end-to-end implementation in which face and voice embeddings are protected by 256-bit BioHash templates [7] under HKDF-derived keys [8], decisions are expressed through calibrated estimates of the original embedding metrics, and every enrollment is issued as a pool of four revocable template sets.
- 2) A dynamic Z-gesture modality: browser-side MediaPipe landmarks [9], a normalized palm trajectory with timing and orientation features, a lenient structural validity gate, dynamic time warping (DTW) [10] with a bounded rotation search, and a keyed orthonormal sequence transform that leaves DTW distances unchanged.
- 3) A fail-closed decision layer: per-modality capture-quality gates that separate capture failures from mismatches, and an all-required fusion policy.
- 4) An evaluation that labels its evidence: public-dataset results for face and voice, including the operating points of the deployed thresholds, and clearly separated synthetic results for the gesture.

II. RELATED WORK

Face verification uses deep embeddings compared by cosine or Euclidean distance; FaceNet [11] and ArcFace [12] are standard training objectives, and MTCNN [13] is a common detector and aligner. Speaker verification relies on neural embeddings; ECAPA-TDNN [14] is a strong text-independent architecture, and VoxCeleb [15] is a standard benchmark.

Behavioral biometrics authenticate how a person acts rather than what they are [16]. In-air and motion gestures have been used for authentication with accelerometers (uWave, DTW-based matching [17]) and with depth cameras for in-air handwriting (KinWrite [18]). These systems store unprotected

[FIGURE 1 PLACEHOLDER — OVERALL SYSTEM ARCHITECTURE]

Lanes: Face · Voice · Z Gesture → keyed template protection (HKDF; BioHash / Q_{uv}) → template store (4 sets) → decision (ALL_REQUIRED)
 solid arrows: enrollment (5 poses, 2 sentences, 3 Z samples); dashed arrows: authentication (one sample per presented modality)
 (diagram to be inserted; see Figure Plan)

Fig. 1. System architecture. Solid arrows: enrollment (5 face poses, 2 sentences, 3 Z samples); dashed arrows: authentication (one sample per presented modality). Only protected templates are stored.

trajectories and are unimodal. Camera-only hand tracking has become practical with MediaPipe Hands [9], which estimates 21 hand landmarks per frame in real time on commodity devices. DTW [10] remains the standard way to compare sequences performed at different speeds.

Fusion can occur at the sensor, feature, score, or decision level [5], [6]; score-level sum rules are simple and effective, but decision rules determine whether one strong modality can override another. Template protection requires revocability, unlinkability, and non-invertibility without destroying accuracy [3], [4]. Cancelable transforms were introduced by Ratha et al. [3]; BioHashing combines a user token with random projection and thresholding [7], but its security degrades when the token (key) is stolen [19]. Unlinkability can be quantified with the framework of Gomez-Barrero et al. [20].

Table I summarizes representative work. Existing protected systems are mostly unimodal, and gesture-authentication systems typically lack template protection. The gap addressed here is an integrated system in which a behavioral camera modality is protected, revoked, and fused with face and voice under one lifecycle, together with an evaluation that states what has and has not been measured.

TABLE I
COMPARISON WITH RELATED WORK

Work	Modality	Method	Protection / fusion	Limitation
Ratha et al. [3]	Generic	Cancelable transforms	Cancelable / —	Concept
Jin et al. [7]	Fingerprint	Random projection + thresholding	BioHash / —	Unimodal; weak if token stolen [19]
Ross & Jain [5]	Multimodal	Fusion levels	— / all levels	No template protection
Liu et al. [17]	Accelerometer gesture	DTW	— / —	Dedicated sensor; unprotected
Tian et al. [18]	In-air handwriting (depth camera)	Trajectory matching	— / —	Depth sensor; unprotected
This work	Face, voice, Z gesture	Embeddings + BioHash; DTW + keyed transform	BioHash / keyed orthonormal; ALL_REQUIRED decision	No real-participant study yet

III. PROPOSED SYSTEM

A. Architecture and Protocol

The system (Fig. 1) consists of a React browser client, a FastAPI backend, and a relational template store. Face images and voice recordings are uploaded as images and WAV files; for the gesture, MediaPipe Hands runs in the browser and only the landmark sequence (never video) is uploaded. Every modality follows the same pattern: capture → quality gate → representation → keyed transformation → protected template; authentication repeats the path with a fresh capture and compares in the protected domain. Table II lists the protocol. Face and voice are required at registration; the gesture is optional. At authentication the user presents one sample of each chosen, enrolled modality; a modality that is not enrolled is rejected before any processing.

B. Face Verification

MTCNN [13] detects the face; a capture is rejected if detection confidence is below 0.90, sharpness (Laplacian variance) below 25, the face occupies less than 15% of the frame, or roll exceeds 20°. The bounding-box crop (160×160) is embedded by InceptionResnetV1 pretrained on VGGFace2 [21] and fine-tuned with ArcFace [12] on LFW identities [22], giving a 512-D L2-normalized embedding. Enrollment guides five poses (front, left, right, up, down), requires at least three valid ones, and forms the centroid

$$e = (\sum_k e_k) / \|\sum_k e_k\|_2, \quad (1)$$

after which the per-pose embeddings are discarded. Authentication embeds one capture.

C. Voice Verification

Recordings are resampled to 16 kHz, silence-trimmed, RMS-normalized, and converted to 80-bin log-mel features over a 4 s window; an ECAPA-TDNN [14] trained with ArcFace yields a 192-D embedding. A capture-quality gate rejects recordings with less than 1.5 s of speech, an estimated SNR below 13 dB, or more than 1% clipped samples; a rejected recording is reported as a capture failure, never as a mismatch. Enrollment records two sentences; their cosine similarity sets a consistency band (≥ 0.75 enrolled, 0.60–0.75 enrolled only after user confirmation, < 0.60 rejected), and the template is built from their normalized centroid (1). Authentication uses exactly one sentence.

TABLE II
PROTOCOL AND CONFIGURATION

Modality	Enrollment	Authentication	Representation / protection	Decision rule
Face	5 guided poses (≥ 3 valid) $\rightarrow$ centroid	1 capture	512-D InceptionResnetV1 / 256-bit BioHash	Estimated cosine ≥ 0.80 (project baseline)
Voice	2 sentences (consistency check) $\rightarrow$ centroid	1 sentence	192-D ECAPA-TDNN / 256-bit BioHash	Estimated Euclidean ≤ 0.75 (project baseline)
Z gesture	3 samples, stored separately	1 gesture	64×13 sequence / keyed orthonormal 13×13	Median DTW ≤ 0.216 (synthetic development value)
Fusion	4 template sets (1 active, 3 standby)	—	—	ALL_REQUIRED over presented modalities

D. Dynamic Z-Gesture Verification

The user draws a Z (left to right, diagonally down-left, left to right) with one hand. The Z has three strokes and two direction changes, making it easy to explain and to validate, while its proportions, curvature, and rhythm vary between people.

Capture and gating. For each video frame the browser records the 21 MediaPipe landmarks (x, y, z) [9] or a null entry. The backend requires ≥ 15 tracked frames, hand presence in $\geq 60\%$ of the span, a tracking rate ≥ 10 frames/s, and 0.5–10 s of movement; coordinates are made isotropic and mirrored back to the user's view. Missing frames are interpolated and a Gaussian kernel in time ($\sigma = 60$ ms) smooths landmark jitter, so the blur is identical at 12 or 30 frames/s. Idle hand before and after the gesture is trimmed: the gesture spans the first to last frame whose palm speed exceeds $\max(0.5 \text{ palm sizes/s}, 0.15 \cdot p90 \text{ speed})$, so pauses inside the Z are kept.

Structural validity. The palm-centre trajectory is resampled to 48 points by arc length, and an exhaustive search over two corner positions accepts the capture if three strokes point right ($\pm 55^\circ$), down-left (90° – 180° , image y downward), and right again, each $\geq 10\%$ of the total chord length, explaining $\geq 55\%$ of the path. This check rejects incomplete, reversed, and mirrored Zs and other shapes; it decides only whether a capture is comparable and never contributes to identity.

Representation. The trimmed sequence is resampled uniformly in time to $T = 64$ points. With palm centre p_t (mean of the wrist and four finger-MCP landmarks),

$$x_t = (p_t - p) / \sqrt{((1/T) \sum_t \|p_t - p\|^2)}, \quad (2)$$

which removes the position and size of the Z. Each point carries 13 features: trajectory x_t (2, weight 1.0), unit direction (2, 0.6), relative speed and its change (2, 0.3 and 0.15), hand orientation wrist $\rightarrow$ middle-MCP (2, 0.3), and five fingertip-to-wrist distances in palm sizes (5, 0.1 each). Speeds are divided by the sample's own mean, so absolute tempo is not a feature.

Matching. For sequences A and B with frame cost $c(i, j) = \|a_i - b_j\|_2$, DTW [10] accumulates

$$D(i, j) = c(i, j) + \min\{D(i-1, j-1), D(i-1, j), D(i, j-1)\} \quad (3)$$

inside a Sakoe–Chiba band of $0.2 \cdot T$ and normalizes by the warping-path length L : $d_{\text{DTW}}(A, B) = D(T, T)/L$. Small orientation differences are handled by rotating the 2-D features

of the live sample by $r \in R = \{0^\circ, \pm 6^\circ, \pm 12^\circ\}$. With three enrolled samples $S_1, \dots, S_3$,

$$d = \text{median}_k [\min_{r \in R} d_{\text{DTW}}(\text{Rot}_r(A), S_k)], \quad (4)$$

accept $\Leftrightarrow d \leq \tau_h = 0.216$.

The median requires the live sample to agree with two of three samples. The three samples are stored separately rather than averaged, because they are not time-aligned. At enrollment, pairwise distances select a medoid and refuse a sample whose distance to the nearer of the other two exceeds both τ_h and twice the distance between those two, so one abnormal sample cannot enter the template. τ_h is a development value (Section IV).

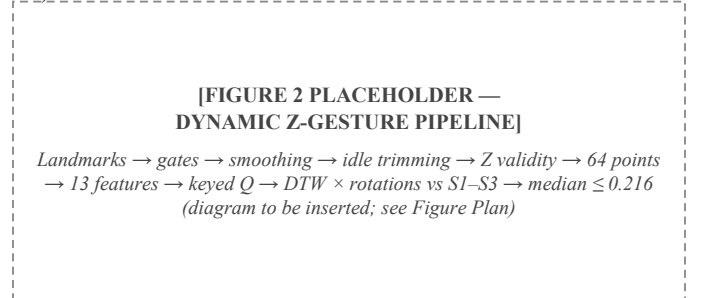


Fig. 2. Dynamic Z-gesture pipeline. The structural check only gates comparability; identity is decided by the median rotation-searched DTW distance to three enrolled samples (0.216: synthetic development threshold).

E. Cancelable Template Protection

Keys are derived per application, user, modality, and key version v with HKDF-SHA256 [8] from a server-side master secret, yielding independent seeds for projection, thresholds, and permutation. For face and voice (Fig. 3), a unit embedding $e \in \mathbb{R}^D$ is projected by a keyed matrix W_v with orthonormal rows (QR of a seeded Gaussian; two blocks for $D = 192$), quantized against keyed thresholds t_v , and permuted:

$$b = \pi_v(1[W_v e > t_v]) \in \{0, 1\}^{256}. \quad (5)$$

Templates are compared by the fraction h of equal bits. Because the probability that a random hyperplane separates two vectors is θ/π [23], h is a monotone function of the embedding angle θ ; the system fits $h = (p_1 \theta + p_2 \theta^2 + p_3 \theta^3)/\pi$ on synthetic pairs and inverts it, so the face rule is an estimated cosine $\hat{c} \geq 0.80$ and the voice rule an estimated Euclidean distance $d' = \sqrt{2 - 2\hat{c}} \leq 0.75$. For the gesture, each weighted frame is multiplied by a keyed orthonormal 13×13 matrix Q_v ; since $\|Q_v a - Q_v b\| = \|a - b\|$, every DTW distance is preserved

exactly. Enrollment produces four template sets (one active, three standby) under consecutive key versions; revocation promotes a standby set for all modalities at once, and an exhausted pool requires re-enrollment. BioHash quantization discards information, but no formal non-invertibility proof is claimed; the gesture transform is invertible by any holder of the key.

**[FIGURE 3 PLACEHOLDER —
CANCELABLE TEMPLATE PROTECTION]**

$HKDF\text{-}SHA256 \rightarrow W_v, t_v, \pi_v \rightarrow 256\text{-bit BioHash (face, voice)}$
 Q_v : keyed orthonormal transform (gesture)
 (diagram to be inserted; see Figure Plan)

Fig. 3. Keyed template protection for face/voice (BioHash) and gesture (orthonormal sequence transform).

F. Multimodal Fusion and Decision

Each modality first decides against its own threshold. For reporting, scores are mapped to a common higher-is-better scale (face $\hat{c}$; voice $1 - d^2/2$; gesture $\text{clip}(1 - d/2\tau_h, -1, 1)$) and averaged with equal weights; the fusion threshold is the mean of the per-modality thresholds on that scale. The decision itself is taken by a policy: ALL_REQUIRED (default) accepts only if every presented modality matches, so a strong face score cannot override a failed voice or gesture; AT_LEAST_TWO and a floor-vetoed WEIGHTED sum are available for comparison. The system fails closed: a capture error or failed quality gate in any presented modality denies the attempt before fusion and is reported as a retry request rather than a mismatch; only repeated high-quality mismatches are flagged as suspicious.

IV. EXPERIMENTAL SETUP

Face. Held-out LFW [22] identities with 2–19 images, disjoint from the 62 fine-tuning identities: 1,618 identities, 4,523 genuine and 80,900 impostor comparisons (50 impostors per identity). The first image is the reference; each probe is templated under the target’s key. This single-reference protocol differs from the deployed five-pose centroid.

Voice. The 24-speaker VoxCeleb1 [15] subset used for training (closed set). Protocols with one or two enrollment utterances and one or two probe utterances are evaluated on the same clips (690 genuine and 15,870 impostor comparisons for the deployed two-enrollment/one-probe case).

Fusion. Chimeric virtual users pair LFW faces with VoxCeleb voices (20 pairings $\times$ 24 users; 72 genuine and 1,656 impostor attempts per pairing), assuming independence; voice uses one enrollment and one probe utterance.

Gesture. No real multi-person gesture dataset exists yet. A generator models per-person style (proportions, slant, stroke curvature, per-stroke speed profile and timing, corner pauses, hand orientation, finger geometry) and per-execution variation (1.5–3.1 s duration, size, position, camera distance, $\pm 6^\circ$ rotation, jitter). τ_h was chosen as the equal-error threshold of the median rule on the development half of 40 synthetic users (seeds disjoint from the test users) and evaluated once on 40 further synthetic users (240 genuine, 9,360 impostor

comparisons). A second synthetic set adds non-uniform tempo change (1.4–3.8 s, altered speed profile and pauses, 12–30 frames/s, a second session); 15 held-out users give 150 genuine and 2,100 impostor comparisons. Live gesture captures were made only during development by one developer.

Metrics. FAR = accepted impostor attempts / impostor attempts; FRR = rejected genuine attempts / genuine attempts; EER is the operating point where FAR = FRR; ROC-AUC is the area under the ROC curve. Latency was measured on an Intel i5-1235U CPU (Windows 11, Python 3.13, no GPU).

V. RESULTS AND DISCUSSION

A. Face and Voice

Table III reports the protected-domain results. Protection costs 1.38 percentage points of face EER (7.12% $\rightarrow$ 8.50%) and 0.12 points of voice EER (2.21% $\rightarrow$ 2.32%). The configured thresholds, however, are conservative: at $\hat{c} \geq 0.80$ the face FAR is 0.0025% but the FRR is 81.5%, despite an AUC of 0.9641. For voice, the protocol matters as much as the threshold: at $d \leq 0.75$, one enrollment and one probe utterance give FRR 24.1%, while the deployed two-sentence enrollment with one-sentence verification gives FRR 12.2%, FAR 0.03%, and EER 1.27%, consistent with the enrollment centroid averaging out utterance-level variation. Noise remains the dominant failure: under the single-utterance protocol at 20 dB SNR, the voice FRR at the same threshold rises to 68.5%. A landmark-aligned face variant, evaluated but not deployed, lowers the protected EER from 8.67% to 7.18% (paired bootstrap 95% CI of the difference -1.90 to -0.69 points).

B. Fusion

On chimeric users, ALL_REQUIRED admits no impostor (FAR 0) but rejects 85.3% of genuine attempts, compounding the face threshold; WEIGHTED lowers the FRR to 59.6% with FAR 0. Score-level averaging of face and voice would reach 1.54% EER (AUC 0.9988) against 8.16% and 2.75% for the individual scores on the same users, indicating complementary errors; it is not the deployed decision rule. These numbers use the earlier single-utterance voice protocol.

C. Dynamic Z Gesture (synthetic)

On the 40 independent synthetic users (Table IV), genuine median distances average 0.129 (95th percentile 0.165) and impostor distances 0.370 (5th percentile 0.243); at $\tau_h = 0.216$ FAR is 1.35% and FRR 0%. With realistic, non-uniform tempo variation the separation remains (EER 3.3%, AUC 0.996), but the FRR at 0.216 rises to 14.7%, showing that τ_h does not transfer to data with more tempo variation. Removing the relative-speed features lowered the held-out EER to 2.6%; resampling by arc length instead of time did not improve separation (3.4%); and fingertip features did not change separation. Development captures by one person agreed: distances rose when the Z was drawn markedly faster or slower, or with a different finger pose, than at enrollment. These results validate the pipeline and expose its sensitivities; they are not real-world error rates.

TABLE III
PROTECTED-TEMPLATE RESULTS ON PUBLIC DATA

Modality / protocol	Comparisons (gen / imp)	EER	ROC-AUC	FAR / FRR at deployed threshold
Face, LFW held-out, single reference	4,523 / 80,900	8.50% (CI 7.77–8.93)	0.9641	0.0025% / 81.5% ($\hat{c} \geq 0.80$)
Face, raw embeddings (reference only)	same	7.12%	0.9682	—
Voice, 1 enroll / 1 probe utterance	727 / 17,273	2.32% (CI 1.95–3.03)	0.9967	—
Voice, 1 / 1 (protocol study)	690 / 15,870	2.31%	—	0.26% / 24.1% ($d \leq 0.75$)
Voice, 2 / 1 (deployed)	690 / 15,870	1.27%	—	0.03% / 12.2%
Fusion, ALL_REQUIRED (chimeric)	72 / 1,656 per pairing $\times$ 20	—	—	0% / 85.3%
Fusion, WEIGHTED (chimeric)	same	—	—	0% / 59.6%
Score-level mean (chimeric, not deployed)	1,440 / 33,120	1.54%	0.9988	—

TABLE IV
Z-GESTURE EVALUATION ON SYNTHETIC USERS

Set / representation	Gen / imp	Genuine mean (p95)	Impostor mean (p5)	EER / ROC-AUC	FAR / FRR at 0.216
Independent set, deployed representation	240 / 9,360	0.129 (0.165)	0.370 (0.243)	0.42% / 0.9999	1.35% / 0%
Tempo stress, deployed representation	150 / 2,100	0.181 (0.236)	0.407 (0.265)	3.3% / 0.9958	0.8% / 14.7%
Tempo stress, no speed features	150 / 2,100	0.156 (0.207)	0.352 (0.230)	2.6% / 0.9971	—
Tempo stress, arc-length + timing	150 / 2,100	0.142 (0.197)	0.319 (0.208)	3.4% / 0.9945	—

D. Revocation, Unlinkability, and Cost

The same face embedding under different set keys gives Hamming similarity 0.499 ± 0.032 ($n = 900$); a revoked template against its replacement gives 0.500 (face) and 0.504 (voice) with no acceptances, while genuine similarity is unchanged after three revocations (face $0.761 \rightarrow 0.766$; voice $0.811 \rightarrow 0.804$). Cross-key linkability $D^{\text{sys}}_{\leftrightarrow}$ [20] is 0.018 for face—small, but above the permutation null floor (0.013)—and 0.020 for voice, indistinguishable from chance. A complete face-and-voice authentication request takes 778 ms on average; BioHash generation takes 43.5 ms and a Hamming comparison 0.006 ms. Backend gesture authentication (feature extraction and 15 DTW comparisons) takes about 31 ms on synthetic input; MediaPipe inference runs on the client.

VI. SECURITY ANALYSIS, LIMITATIONS, AND FUTURE WORK

Implemented protections. Only protected templates, key versions, and non-biometric metadata are stored; video, audio, and embeddings are discarded after processing. Templates are revocable per set and keyed per application, user, and modality. Decisions are fail-closed, capture failures are not counted as mismatches, repeated high-quality mismatches are flagged, uploads are type- and size-restricted, and every attempt is audited.

Limitations. (i) Key compromise: all keys derive from one master secret; with it, BioHash loses much of its protection [19], and the gesture templates can be inverted. (ii) No presentation-attack detection or liveness test is implemented; face replay, voice replay or synthesis, and gesture imitation are untested, and the backend trusts client-supplied landmarks, which an attacker could fabricate. (iii) No real-participant evaluation exists for any modality: the voice results are closed-set and on clean clips, the face results use a single reference image, the fusion results assume independent

chimeric users, and the gesture threshold is synthetic. (iv) The face and voice thresholds are fixed project baselines, not tuned operating points. (v) Demographic effects were not evaluated.

Future work. A consented multi-session pilot (at least ten participants with genuine and impostor gestures) to select the gesture representation and threshold on development data and report them once on held-out participants; re-tuning the face and voice operating points; presentation-attack detection [24]; server-side verification of landmark provenance; hardware-backed key storage; and a non-invertible sequence transform for the gesture.

VII. CONCLUSION

This paper presented a cancelable multimodal verification system that combines face, voice, and a dynamic Z hand gesture, stores only keyed templates grouped into revocable sets, and decides with a fail-closed all-required policy. On public data, protection costs 1.4 EER percentage points for face and 0.1 for voice, revoked templates agree with their replacements only at chance level, and the deployed two-sentence voice enrollment halves the single-enrollment false-rejection rate. The results also show that the fixed baseline thresholds reject most genuine face attempts and that the gesture threshold, derived from synthetic users, does not transfer to data with realistic tempo variation. The system is a research prototype, not a production deployment; calibrated thresholds, liveness detection, and a real participant study are required before its error rates can be stated for real users.

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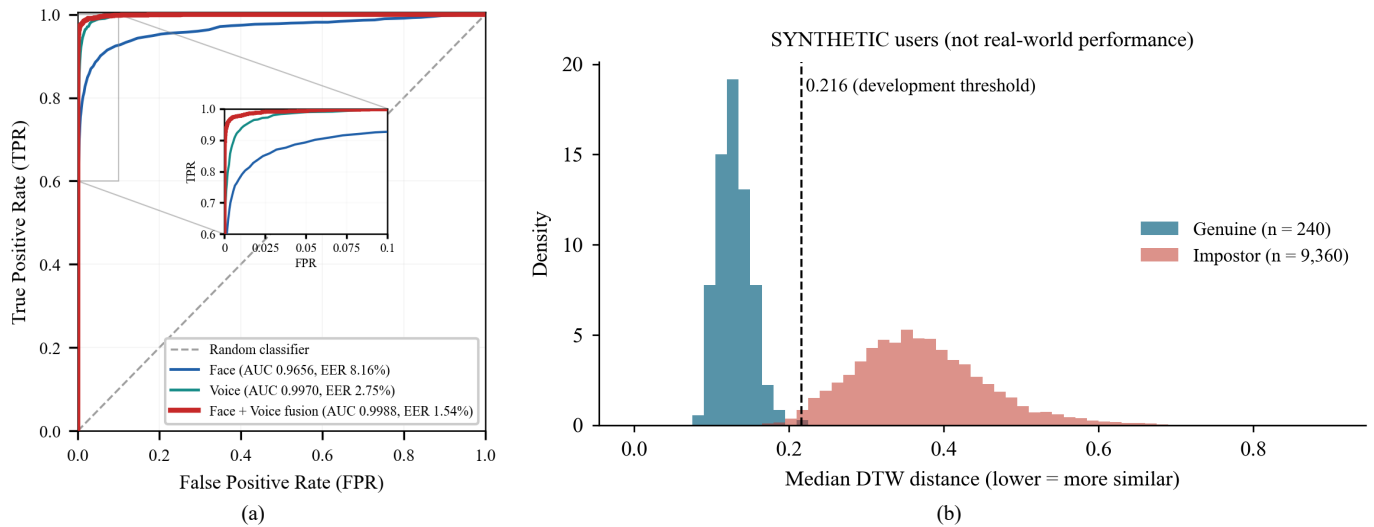


Fig. 4. (a) ROC of protected face and voice templates on chimeric users; (b) genuine/impostor median-DTW distances for synthetic gesture users (line: 0.216). Panel (b) is not real-world performance.

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